Urban Policies to Combat Climate Change: Assessing the Impact of Congestion Pricing and Low-Emission Zones using Machine Learning Techniques

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Executive Summary

Abstract content goes here. This section should provide a concise summary of the research question, methodology, key findings, and implications of the study. It should be clear and informative, allowing readers to quickly understand the purpose and results of the paper.

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Replication files are available in the GitHub repository for this project.

Table 1: Cross-fitted RMSE for predicting Nuisance Parameters

Model	OLS - Basic	L1 (Lasso / Logit L1)	L2 (Ridge / Logit L2)	Elastic Net
RMSE Y (Outcome)	0.5539	0.5250	0.5500	0.5250
RMSE D (CP)	0.3470	0.3058	0.3077	0.3067
RMSE D (LEZ)	0.3315	0.2822	0.2826	0.2846
RMSE D (CP $\times$ LEZ)	0.3009	0.2522	0.2563	0.2541

1 Introduction

2 Empirical Strategy and Identification

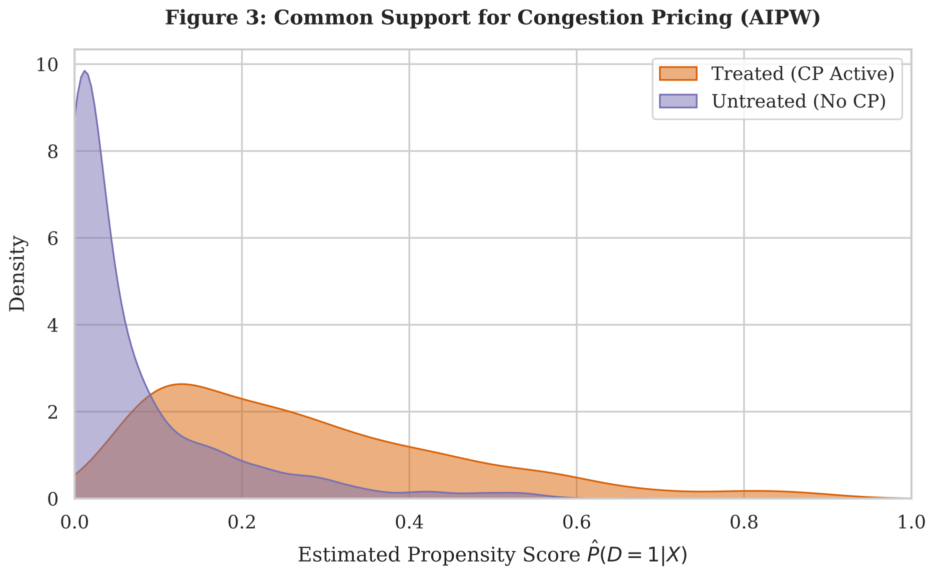

Figure 1: Common Support for Congestion Pricing (AIPW)

Table 2: Double Machine Learning Estimates of Urban Climate Policies (Categorical Regime)

Policy	(1) OLS Baseline	(2) AIPW (Lasso)	(3) Causal Fores
Panel A: Global Average Treatment Effect (ATE)			
Congestion Pricing (CP)	-0.9505^{***} ($p = 0.000$)	-0.2569^{***} ($p = 0.000$)	-0.3530^{***} ($p = 0.0$
Low Emission Zone (LEZ)	-0.4404^{***} ($p = 0.001$)	-0.0850^{***} ($p = 0.000$)	-0.0780^{***} ($p = 0.0$
Synergy (CP × LEZ)	0.2023 ($p = 0.458$)	-1.0337^{***} ($p = 0.000$)	-0.7282^{***} ($p = 0.0$
Panel B: Average Treatment Effect on the Treated (ATT)			
Congestion Pricing (CP)	-1.0486^{***} ($p = 0.000$)	-0.3006^{***} ($p = 0.000$)	-0.3658^{***} ($p = 0.0$
Low Emission Zone (LEZ)	-0.4016^{***} ($p = 0.001$)	-0.0957^{***} ($p = 0.000$)	-0.0999^{***} ($p = 0.0$
Synergy (CP × LEZ)	-0.0660 ($p = 0.847$)	-1.1911^{***} ($p = 0.000$)	-1.0307^{***} ($p = 0.0$
Panel C: Group ATT (Heterogeneity by City Type)			
Cluster 1: Dense Metropolis			
Congestion Pricing (CP)	-1.2302^{***} ($p = 0.000$)	-0.3815^{***} ($p = 0.000$)	-0.3896^{***} ($p = 0.0$
Low Emission Zone (LEZ)	-0.2842^{*} ($p = 0.082$)	-0.1279^{***} ($p = 0.000$)	-0.1661^{***} ($p = 0.0$
Synergy (CP × LEZ)	-0.2836 ($p = 0.533$)	-1.3187^{***} ($p = 0.000$)	-1.2759^{***} ($p = 0.0$
Cluster 0: Sprawling Cities			
Congestion Pricing (CP)	-0.7732^{***} ($p = 0.000$)	-0.1779^{***} ($p = 0.000$)	-0.3298^{***} ($p = 0.0$
Low Emission Zone (LEZ)	-0.5394^{***} ($p = 0.003$)	-0.0578^{***} ($p = 0.008$)	-0.0222 ($p = 0.27$
Synergy (CP × LEZ)	0.5103 ($p = 0.133$)	-0.8529^{***} ($p = 0.000$)	-0.3809^{***} ($p = 0.0$

3 Proposed Estimator

4 Theoretical Properties

5 Data Preparation

6 Results

6.1 Nuisance Parameter Evaluation

6.2 Main Causal Estimates

Notes: p-values in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. 'Failed' estimation bounds denote overlap violations.

6.3 Spatial Heterogeneity

7 Sensitivity Analysis

Notes: p-values in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

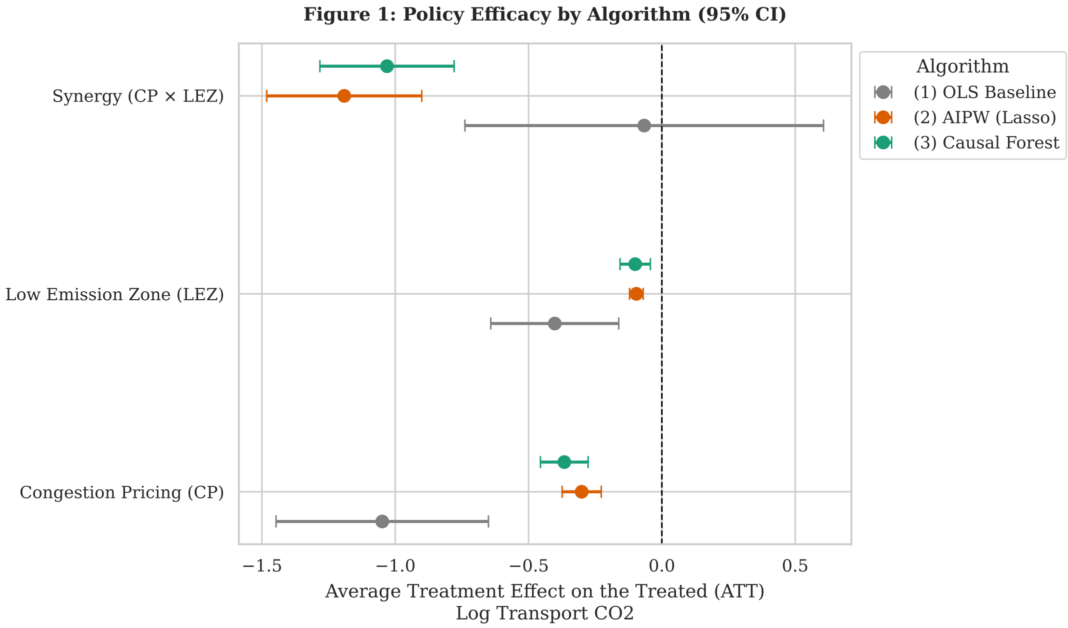

Figure 2: Policy Efficacy by Algorithm (95% CI)

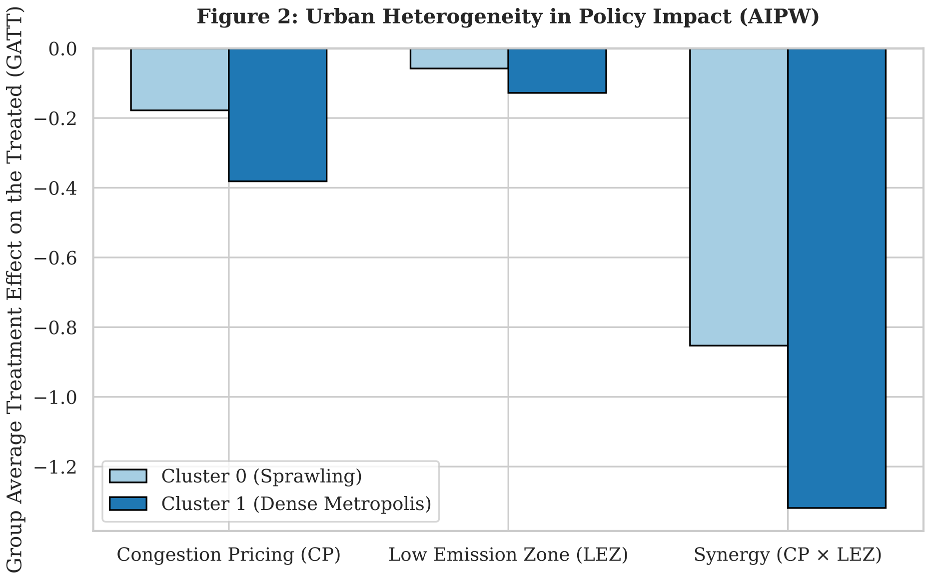

Figure 3: Urban Heterogeneity in Policy Impact (AIPW)

Table 3: Falsification Tests: Estimated Effects on Placebo Outcomes (Global ATT)

Model	AIPW (Lasso)		
Policy Decoy Outcome	Congestion Pricing (CP)	Low Emission Zone (LEZ)	Synergy (CP × LE
Library Count	-0.2060^{***} ($p = 0.000$)	0.0348 ($p = 0.360$)	-0.0142 ($p = 0.705$
Streetlight Density	4.3843^{***} ($p = 0.001$)	-3.2133^{***} ($p = 0.000$)	-1.5590^{*} ($p = 0.08$
Fountain Count	-0.2088 ($p = 0.176$)	0.6461^{***} ($p = 0.000$)	0.3165^{**} ($p = 0.014$
Bench Count (per cap)	0.1338 ($p = 0.299$)	-0.5187^{***} ($p = 0.000$)	-0.5976^{***} ($p = 0.00$
Flagpole Count	-0.2321^{*} ($p = 0.075$)	-0.3132^{***} ($p = 0.006$)	-0.1669 ($p = 0.176$
Sister City Count	0.4016^{***} ($p = 0.002$)	0.4540^{***} ($p = 0.000$)	0.6058^{***} ($p = 0.00$

8 Interpretation and Discussion

Appendix

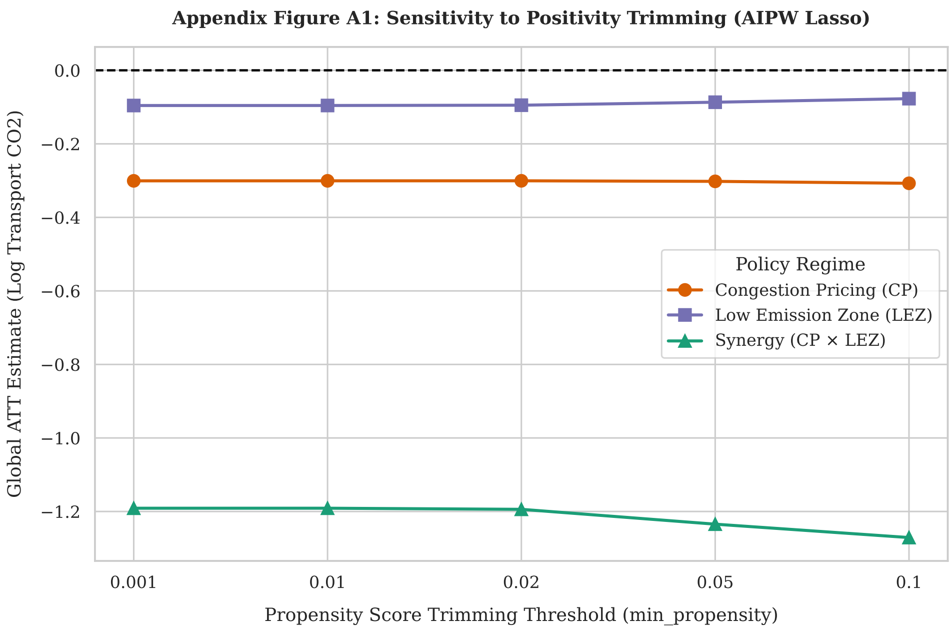

Figure 4: Sensitivity to Positivity Trimming

Declaration of Autorship

I confirm that this report is based on my own work. In preparing this report, I have not received any help from another human nor have I discussed any aspects of the empirical project with others.

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